Filter Summary Report: CG,TIA,Automated,NO,L,simple,Z2,Z3,Z4

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Contents

1 Examined $H(z)$ for CG TIA Automated NO L simple Z2 Z3 Z4: $\frac{Z_2Z_3Z_4g_m+Z_3Z_4}{2Z_2Z_3g_m+Z_2Z_4g_m+2Z_3+Z_4}$

$$H(z) = \frac{Z_2Z_3Z_4g_m + Z_3Z_4}{2Z_2Z_3g_m + Z_2Z_4g_m + 2Z_3 + Z_4}$$

2 AP

3 BP

4 BS

5 GE

6 HP

7 LP

8 X-INVALID-NUMER

8.1 X-INVALID-NUMER-1 $Z(s) = \left(\infty,\ R_2,\ \frac{R_3}{C_3R_3s+1},\ R_4 + \frac{1}{C_4s},\ \infty,\ \infty\right)$

$$H(s) = \frac{R_2R_3g_m + R_3 + s\left(C_4R_2R_3R_4g_m + C_4R_3R_4\right)}{R_2g_m + s^2\left(C_3C_4R_2R_3R_4g_m + C_3C_4R_3R_4\right) + s\left(C_3R_2R_3g_m + C_3R_3 + 2C_4R_2R_3g_m + C_4R_2R_4g_m + 2C_4R_3 + C_4R_4\right) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_3}\sqrt{C_4}\sqrt{R_3}\sqrt{R_4}}{C_3R_3+2C_4R_3+C_4R_4}$
wo: $\frac{1}{\sqrt{C_3}\sqrt{C_4}\sqrt{R_3}\sqrt{R_4}}$
bandwidth: $\frac{C_3R_3+2C_4R_3+C_4R_4}{C_3C_4R_3R_4}$
K-LP: R_3
K-HP: 0
K-BP: $\frac{C_4R_3R_4}{C_3R_3+2C_4R_3+C_4R_4}$
Qz: None
Wz: None

8.2 X-INVALID-NUMER-2 $Z(s) = \left(\infty,\ R_2,\ R_3 + \frac{1}{C_3s},\ \frac{R_4}{C_4R_4s+1},\ \infty,\ \infty\right)$

$$H(s) = \frac{R_2R_4g_m + R_4 + s\left(C_3R_2R_3R_4g_m + C_3R_3R_4\right)}{2R_2g_m + s^2\left(2C_3C_4R_2R_3R_4g_m + 2C_3C_4R_3R_4\right) + s\left(2C_3R_2R_3g_m + C_3R_2R_4g_m + 2C_3R_3 + C_3R_4 + 2C_4R_2R_4g_m + 2C_4R_4\right) + 2}$$

Parameters:

Q: $\frac{2\sqrt{C_3}\sqrt{C_4}\sqrt{R_3}\sqrt{R_4}}{2C_3R_3+C_3R_4+2C_4R_4}$
wo: $\frac{1}{\sqrt{C_3}\sqrt{C_4}\sqrt{R_3}\sqrt{R_4}}$
bandwidth: $\frac{2C_3R_3+C_3R_4+2C_4R_4}{2C_3C_4R_3R_4}$
K-LP: $\frac{R_4}{2}$
K-HP: 0
K-BP: $\frac{C_3R_3R_4}{2C_3R_3+C_3R_4+2C_4R_4}$
Qz: None
Wz: None

8.3 X-INVALID-NUMER-3 $Z(s) = \left(\infty,\ \frac{1}{C_2 s},\ R_3,\ \frac{1}{C_4 s},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_2 R_3 s + R_3 g_m}{2C_2C_4R_3s^2 + g_m + s\left(C_2 + 2C_4R_3g_m\right)}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_2}\sqrt{C_4}\sqrt{R_3}\sqrt{g_m}}{C_2+2C_4R_3g_m}$
wo: $\frac{\sqrt{2}\sqrt{g_m}}{2\sqrt{C_2}\sqrt{C_4}\sqrt{R_3}}$
bandwidth: $\frac{C_2+2C_4R_3g_m}{2C_2C_4R_3}$
K-LP: R_3
K-HP: 0
K-BP: $\frac{C_2R_3}{C_2+2C_4R_3g_m}$
Qz: None
Wz: None

8.4 X-INVALID-NUMER-4 $Z(s) = \left(\infty,\ \frac{1}{C_2 s},\ R_3,\ \frac{R_4}{C_4R_4s+1},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_2R_3R_4s + R_3R_4g_m}{2C_2C_4R_3R_4s^2 + 2R_3g_m + R_4g_m + s\left(2C_2R_3 + C_2R_4 + 2C_4R_3R_4g_m\right)}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_2}\sqrt{C_4}\sqrt{R_3}\sqrt{R_4}\sqrt{g_m}\sqrt{2R_3+R_4}}{2C_2R_3+C_2R_4+2C_4R_3R_4g_m}$
wo: $\frac{\sqrt{2}\sqrt{2R_3g_m+R_4g_m}}{2\sqrt{C_2}\sqrt{C_4}\sqrt{R_3}\sqrt{R_4}}$
bandwidth: $\frac{\sqrt{2R_3g_m+R_4g_m}(2C_2R_3+C_2R_4+2C_4R_3R_4g_m)}{2C_2C_4R_3R_4\sqrt{g_m}\sqrt{2R_3+R_4}}$
K-LP: $\frac{R_3R_4}{2R_3+R_4}$
K-HP: 0
K-BP: $\frac{C_2R_3R_4}{2C_2R_3+C_2R_4+2C_4R_3R_4g_m}$
Qz: None
Wz: None

8.5 X-INVALID-NUMER-5 $Z(s) = \left(\infty,\ \frac{1}{C_2 s},\ \frac{1}{C_3 s},\ R_4,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_2R_4s + R_4g_m}{C_2C_3R_4s^2 + 2g_m + s\left(2C_2 + C_3R_4g_m\right)}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_2}\sqrt{C_3}\sqrt{R_4}\sqrt{g_m}}{2C_2+C_3R_4g_m}$
wo: $\frac{\sqrt{2}\sqrt{g_m}}{\sqrt{C_2}\sqrt{C_3}\sqrt{R_4}}$
bandwidth: $\frac{2C_2+C_3R_4g_m}{C_2C_3R_4}$
K-LP: $\frac{R_4}{2}$
K-HP: 0
K-BP: $\frac{C_2R_4}{2C_2+C_3R_4g_m}$
Qz: None
Wz: None

8.6 X-INVALID-NUMER-6 $Z(s) = \left(\infty,\ \frac{1}{C_2 s},\ \frac{1}{C_3 s},\ \frac{R_4}{C_4R_4s+1},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_2R_4s + R_4g_m}{2g_m + s^2\left(C_2C_3R_4 + 2C_2C_4R_4\right) + s\left(2C_2 + C_3R_4g_m + 2C_4R_4g_m\right)}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_2}\sqrt{R_4}\sqrt{g_m}\sqrt{C_3+2C_4}}{2C_2+C_3R_4g_m+2C_4R_4g_m}$
wo: $\frac{\sqrt{2}\sqrt{g_m}}{\sqrt{C_2C_3R_4+2C_2C_4R_4}}$

bandwidth: $\frac{2C_2+C_3R_4g_m+2C_4R_4g_m}{\sqrt{C_2}\sqrt{R_4}\sqrt{C_3+2C_4}\sqrt{C_2C_3R_4+2C_2C_4R_4}}$
K-LP: $\frac{R_4}{2}$
K-HP: 0
K-BP: $\frac{C_2R_4}{2C_2+C_3R_4g_m+2C_4R_4g_m}$
Qz: None
Wz: None

8.7 X-INVALID-NUMER-7 $Z(s) = \left(\infty,\ \frac{1}{C_2s},\ \frac{R_3}{C_3R_3s+1},\ R_4,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_2R_3R_4s + R_3R_4g_m}{C_2C_3R_3R_4s^2 + 2R_3g_m + R_4g_m + s\left(2C_2R_3 + C_2R_4 + C_3R_3R_4g_m\right)}$$

Parameters:

Q: $\frac{\sqrt{C_2}\sqrt{C_3}\sqrt{R_3}\sqrt{R_4}\sqrt{g_m}\sqrt{2R_3+R_4}}{2C_2R_3+C_2R_4+C_3R_3R_4g_m}$
wo: $\frac{\sqrt{2R_3g_m+R_4g_m}}{\sqrt{C_2}\sqrt{C_3}\sqrt{R_3}\sqrt{R_4}}$
bandwidth: $\frac{\sqrt{2R_3g_m+R_4g_m}(2C_2R_3+C_2R_4+C_3R_3R_4g_m)}{C_2C_3R_3R_4\sqrt{g_m}\sqrt{2R_3+R_4}}$
K-LP: $\frac{R_3R_4}{2R_3+R_4}$
K-HP: 0
K-BP: $\frac{C_2R_3R_4}{2C_2R_3+C_2R_4+C_3R_3R_4g_m}$
Qz: None
Wz: None

8.8 X-INVALID-NUMER-8 $Z(s) = \left(\infty,\ \frac{1}{C_2s},\ \frac{R_3}{C_3R_3s+1},\ \frac{1}{C_4s},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_2R_3s + R_3g_m}{g_m + s^2\left(C_2C_3R_3 + 2C_2C_4R_3\right) + s\left(C_2 + C_3R_3g_m + 2C_4R_3g_m\right)}$$

Parameters:

Q: $\frac{\sqrt{C_2}\sqrt{R_3}\sqrt{g_m}\sqrt{C_3+2C_4}}{C_2+C_3R_3g_m+2C_4R_3g_m}$
wo: $\frac{\sqrt{g_m}}{\sqrt{C_2C_3R_3+2C_2C_4R_3}}$
bandwidth: $\frac{C_2+C_3R_3g_m+2C_4R_3g_m}{\sqrt{C_2}\sqrt{R_3}\sqrt{C_3+2C_4}\sqrt{C_2C_3R_3+2C_2C_4R_3}}$
K-LP: R_3
K-HP: 0
K-BP: $\frac{C_2R_3}{C_2+C_3R_3g_m+2C_4R_3g_m}$
Qz: None
Wz: None

8.9 X-INVALID-NUMER-9 $Z(s) = \left(\infty,\ \frac{1}{C_2s},\ \frac{R_3}{C_3R_3s+1},\ \frac{R_4}{C_4R_4s+1},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_2R_3R_4s + R_3R_4g_m}{2R_3g_m + R_4g_m + s^2\left(C_2C_3R_3R_4 + 2C_2C_4R_3R_4\right) + s\left(2C_2R_3 + C_2R_4 + C_3R_3R_4g_m + 2C_4R_3R_4g_m\right)}$$

Parameters:

Q: $\frac{\sqrt{C_2}\sqrt{R_3}\sqrt{R_4}\sqrt{g_m}\sqrt{C_3+2C_4}\sqrt{2R_3+R_4}}{2C_2R_3+C_2R_4+C_3R_3R_4g_m+2C_4R_3R_4g_m}$
wo: $\frac{\sqrt{2R_3g_m+R_4g_m}}{\sqrt{C_2C_3R_3R_4+2C_2C_4R_3R_4}}$
bandwidth: $\frac{\sqrt{2R_3g_m+R_4g_m}(2C_2R_3+C_2R_4+C_3R_3R_4g_m+2C_4R_3R_4g_m)}{\sqrt{C_2}\sqrt{R_3}\sqrt{R_4}\sqrt{g_m}\sqrt{C_3+2C_4}\sqrt{2R_3+R_4}\sqrt{C_2C_3R_3R_4+2C_2C_4R_3R_4}}$
K-LP: $\frac{R_3R_4}{2R_3+R_4}$
K-HP: 0
K-BP: $\frac{C_2R_3R_4}{2C_2R_3+C_2R_4+C_3R_3R_4g_m+2C_4R_3R_4g_m}$
Qz: None
Wz: None

8.10 X-INVALID-NUMER-10 $Z(s) = \left(\infty,\ \frac{R_2}{C_2R_2s+1},\ R_3,\ \frac{1}{C_4s},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_2R_2R_3s + R_2R_3g_m + R_3}{2C_2C_4R_2R_3s^2 + R_2g_m + s\left(C_2R_2 + 2C_4R_2R_3g_m + 2C_4R_3\right) + 1}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_2}\sqrt{C_4}\sqrt{R_2}\sqrt{R_3}\sqrt{R_2g_m+1}}{C_2R_2+2C_4R_2R_3g_m+2C_4R_3}$
wo: $\frac{\sqrt{2}\sqrt{R_2g_m+1}}{2\sqrt{C_2}\sqrt{C_4}\sqrt{R_2}\sqrt{R_3}}$
bandwidth: $\frac{C_2R_2+2C_4R_2R_3g_m+2C_4R_3}{2C_2C_4R_2R_3}$
K-LP: R_3
K-HP: 0
K-BP: $\frac{C_2R_2R_3}{C_2R_2+2C_4R_2R_3g_m+2C_4R_3}$
Qz: None
Wz: None

8.11 X-INVALID-NUMER-11 $Z(s) = \left(\infty,\ \frac{R_2}{C_2R_2s+1},\ R_3,\ \frac{R_4}{C_4R_4s+1},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_2R_2R_3R_4s + R_2R_3R_4g_m + R_3R_4}{2C_2C_4R_2R_3R_4s^2 + 2R_2R_3g_m + R_2R_4g_m + 2R_3 + R_4 + s\left(2C_2R_2R_3 + C_2R_2R_4 + 2C_4R_2R_3R_4g_m + 2C_4R_3R_4\right)}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_2}\sqrt{C_4}\sqrt{R_2}\sqrt{R_3}\sqrt{R_4}\sqrt{2R_2R_3g_m+R_2R_4g_m+2R_3+R_4}}{2C_2R_2R_3+C_2R_2R_4+2C_4R_2R_3R_4g_m+2C_4R_3R_4}$
wo: $\frac{\sqrt{2}\sqrt{2R_2R_3g_m+R_2R_4g_m+2R_3+R_4}}{2\sqrt{C_2}\sqrt{C_4}\sqrt{R_2}\sqrt{R_3}\sqrt{R_4}}$
bandwidth: $\frac{2C_2R_2R_3+C_2R_2R_4+2C_4R_2R_3R_4g_m+2C_4R_3R_4}{2C_2C_4R_2R_3R_4}$
K-LP: $\frac{R_3R_4}{2R_3+R_4}$
K-HP: 0
K-BP: $\frac{C_2R_2R_3R_4}{2C_2R_2R_3+C_2R_2R_4+2C_4R_2R_3R_4g_m+2C_4R_3R_4}$
Qz: None
Wz: None

8.12 X-INVALID-NUMER-12 $Z(s) = \left(\infty,\ \frac{R_2}{C_2R_2s+1},\ \frac{1}{C_3s},\ R_4,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_2R_2R_4s + R_2R_4g_m + R_4}{C_2C_3R_2R_4s^2 + 2R_2g_m + s\left(2C_2R_2 + C_3R_2R_4g_m + C_3R_4\right) + 2}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_2}\sqrt{C_3}\sqrt{R_2}\sqrt{R_4}\sqrt{R_2g_m+1}}{2C_2R_2+C_3R_2R_4g_m+C_3R_4}$
wo: $\frac{\sqrt{2R_2g_m+2}}{\sqrt{C_2}\sqrt{C_3}\sqrt{R_2}\sqrt{R_4}}$
bandwidth: $\frac{\sqrt{2}\sqrt{2R_2g_m+2}(2C_2R_2+C_3R_2R_4g_m+C_3R_4)}{2C_2C_3R_2R_4\sqrt{R_2g_m+1}}$
K-LP: $\frac{R_4}{2}$
K-HP: 0
K-BP: $\frac{C_2R_2R_4}{2C_2R_2+C_3R_2R_4g_m+C_3R_4}$
Qz: None
Wz: None

8.13 X-INVALID-NUMER-13 $Z(s) = \left(\infty,\ \frac{R_2}{C_2R_2s+1},\ \frac{1}{C_3s},\ \frac{R_4}{C_4R_4s+1},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_2R_2R_4s + R_2R_4g_m + R_4}{2R_2g_m + s^2\left(C_2C_3R_2R_4 + 2C_2C_4R_2R_4\right) + s\left(2C_2R_2 + C_3R_2R_4g_m + C_3R_4 + 2C_4R_2R_4g_m + 2C_4R_4\right) + 2}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_2}\sqrt{R_2}\sqrt{R_4}\sqrt{C_3+2C_4}\sqrt{R_2g_m+1}}{2C_2R_2+C_3R_2R_4g_m+C_3R_4+2C_4R_2R_4g_m+2C_4R_4}$
wo: $\frac{\sqrt{2R_2g_m+2}}{\sqrt{C_2C_3R_2R_4+2C_2C_4R_2R_4}}$
bandwidth: $\frac{\sqrt{2}\sqrt{2R_2g_m+2}(2C_2R_2+C_3R_2R_4g_m+C_3R_4+2C_4R_2R_4g_m+2C_4R_4)}{2\sqrt{C_2}\sqrt{R_2}\sqrt{R_4}\sqrt{C_3+2C_4}\sqrt{R_2g_m+1}\sqrt{C_2C_3R_2R_4+2C_2C_4R_2R_4}}$

K-LP: $\frac{R_4}{2}$
K-HP: 0
K-BP: $\frac{C_2R_2R_4}{2C_2R_2+C_3R_2R_4g_m+C_3R_4+2C_4R_2R_4g_m+2C_4R_4}$
Qz: None
Wz: None

8.14 X-INVALID-NUMER-14 $Z(s) = \left(\infty,\ \frac{R_2}{C_2R_2s+1},\ \frac{R_3}{C_3R_3s+1},\ R_4,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_2R_2R_3R_4s + R_2R_3R_4g_m + R_3R_4}{C_2C_3R_2R_3R_4s^2 + 2R_2R_3g_m + R_2R_4g_m + 2R_3 + R_4 + s\left(2C_2R_2R_3 + C_2R_2R_4 + C_3R_2R_3R_4g_m + C_3R_3R_4\right)}$$

Parameters:

Q: $\frac{\sqrt{C_2}\sqrt{C_3}\sqrt{R_2}\sqrt{R_3}\sqrt{R_4}\sqrt{2R_2R_3g_m+R_2R_4g_m+2R_3+R_4}}{2C_2R_2R_3+C_2R_2R_4+C_3R_2R_3R_4g_m+C_3R_3R_4}$
wo: $\frac{\sqrt{2R_2R_3g_m+R_2R_4g_m+2R_3+R_4}}{\sqrt{C_2}\sqrt{C_3}\sqrt{R_2}\sqrt{R_3}\sqrt{R_4}}$
bandwidth: $\frac{2C_2R_2R_3+C_2R_2R_4+C_3R_2R_3R_4g_m+C_3R_3R_4}{C_2C_3R_2R_3R_4}$
K-LP: $\frac{R_3R_4}{2R_3+R_4}$
K-HP: 0
K-BP: $\frac{C_2R_2R_3R_4}{2C_2R_2R_3+C_2R_2R_4+C_3R_2R_3R_4g_m+C_3R_3R_4}$
Qz: None
Wz: None

8.15 X-INVALID-NUMER-15 $Z(s) = \left(\infty,\ \frac{R_2}{C_2R_2s+1},\ \frac{R_3}{C_3R_3s+1},\ \frac{1}{C_4s},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_2R_2R_3s + R_2R_3g_m + R_3}{R_2g_m + s^2\left(C_2C_3R_2R_3 + 2C_2C_4R_2R_3\right) + s\left(C_2R_2 + C_3R_2R_3g_m + C_3R_3 + 2C_4R_2R_3g_m + 2C_4R_3\right) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_2}\sqrt{R_2}\sqrt{R_3}\sqrt{C_3+2C_4}\sqrt{R_2g_m+1}}{C_2R_2+C_3R_2R_3g_m+C_3R_3+2C_4R_2R_3g_m+2C_4R_3}$
wo: $\frac{\sqrt{R_2g_m+1}}{\sqrt{C_2C_3R_2R_3+2C_2C_4R_2R_3}}$
bandwidth: $\frac{C_2R_2+C_3R_2R_3g_m+C_3R_3+2C_4R_2R_3g_m+2C_4R_3}{\sqrt{C_2}\sqrt{R_2}\sqrt{R_3}\sqrt{C_3+2C_4}\sqrt{C_2C_3R_2R_3+2C_2C_4R_2R_3}}$
K-LP: R_3
K-HP: 0
K-BP: $\frac{C_2R_2R_3}{C_2R_2+C_3R_2R_3g_m+C_3R_3+2C_4R_2R_3g_m+2C_4R_3}$
Qz: None
Wz: None

8.16 X-INVALID-NUMER-16 $Z(s) = \left(\infty,\ \frac{R_2}{C_2R_2s+1},\ \frac{R_3}{C_3R_3s+1},\ \frac{R_4}{C_4R_4s+1},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_2R_2R_3R_4s + R_2R_3R_4g_m + R_3R_4}{2R_2R_3g_m + R_2R_4g_m + 2R_3 + R_4 + s^2\left(C_2C_3R_2R_3R_4 + 2C_2C_4R_2R_3R_4\right) + s\left(2C_2R_2R_3 + C_2R_2R_4 + C_3R_2R_3R_4g_m + C_3R_3R_4 + 2C_4R_2R_3R_4g_m + 2C_4R_3R_4\right)}$$

Parameters:

Q: $\frac{\sqrt{C_2}\sqrt{R_2}\sqrt{R_3}\sqrt{R_4}\sqrt{C_3+2C_4}\sqrt{2R_2R_3g_m+R_2R_4g_m+2R_3+R_4}}{2C_2R_2R_3+C_2R_2R_4+C_3R_2R_3R_4g_m+C_3R_3R_4+2C_4R_2R_3R_4g_m+2C_4R_3R_4}$
wo: $\frac{\sqrt{2R_2R_3g_m+R_2R_4g_m+2R_3+R_4}}{\sqrt{C_2C_3R_2R_3R_4+2C_2C_4R_2R_3R_4}}$
bandwidth: $\frac{2C_2R_2R_3+C_2R_2R_4+C_3R_2R_3R_4g_m+C_3R_3R_4+2C_4R_2R_3R_4g_m+2C_4R_3R_4}{\sqrt{C_2}\sqrt{R_2}\sqrt{R_3}\sqrt{R_4}\sqrt{C_3+2C_4}\sqrt{C_2C_3R_2R_3R_4+2C_2C_4R_2R_3R_4}}$
K-LP: $\frac{R_3R_4}{2R_3+R_4}$
K-HP: 0
K-BP: $\frac{C_2R_2R_3R_4}{2C_2R_2R_3+C_2R_2R_4+C_3R_2R_3R_4g_m+C_3R_3R_4+2C_4R_2R_3R_4g_m+2C_4R_3R_4}$
Qz: None
Wz: None

8.17 X-INVALID-NUMER-17 $Z(s) = \left(\infty,\ R_2 + \frac{1}{C_2 s},\ R_3,\ \frac{1}{C_4 s},\ \infty,\ \infty\right)$

$$H(s) = \frac{R_3 g_m + s\left(C_2 R_2 R_3 g_m + C_2 R_3\right)}{g_m + s^2\left(2C_2C_4R_2R_3g_m + 2C_2C_4R_3\right) + s\left(C_2R_2g_m + C_2 + 2C_4R_3g_m\right)}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_2}\sqrt{C_4}R_2\sqrt{R_3}g_m\sqrt{\frac{g_m}{R_2g_m+1}}+\sqrt{2}\sqrt{C_2}\sqrt{C_4}\sqrt{R_3}\sqrt{\frac{g_m}{R_2g_m+1}}}{C_2R_2g_m+C_2+2C_4R_3g_m}$
wo: $\sqrt{\frac{g_m}{2C_2C_4R_2R_3g_m+2C_2C_4R_3}}$
bandwidth: $\frac{\sqrt{\frac{g_m}{2C_2C_4R_2R_3g_m+2C_2C_4R_3}}\left(C_2R_2g_m+C_2+2C_4R_3g_m\right)}{\sqrt{2}\sqrt{C_2}\sqrt{C_4}R_2\sqrt{R_3}g_m\sqrt{\frac{g_m}{R_2g_m+1}}+\sqrt{2}\sqrt{C_2}\sqrt{C_4}\sqrt{R_3}\sqrt{\frac{g_m}{R_2g_m+1}}}$
K-LP: R_3
K-HP: 0
K-BP: $\frac{C_2R_2R_3g_m+C_2R_3}{C_2R_2g_m+C_2+2C_4R_3g_m}$
Qz: None
Wz: None

8.18 X-INVALID-NUMER-18 $Z(s) = \left(\infty,\ R_2 + \frac{1}{C_2 s},\ R_3,\ \frac{R_4}{C_4R_4s+1},\ \infty,\ \infty\right)$

$$H(s) = \frac{R_3R_4g_m + s\left(C_2R_2R_3R_4g_m + C_2R_3R_4\right)}{2R_3g_m + R_4g_m + s^2\left(2C_2C_4R_2R_3R_4g_m + 2C_2C_4R_3R_4\right) + s\left(2C_2R_2R_3g_m + C_2R_2R_4g_m + 2C_2R_3 + C_2R_4 + 2C_4R_3R_4g_m\right)}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_2}\sqrt{C_4}R_2\sqrt{R_3}\sqrt{R_4}g_m\sqrt{\frac{g_m}{R_2g_m+1}}\sqrt{2R_3+R_4}+\sqrt{2}\sqrt{C_2}\sqrt{C_4}\sqrt{R_3}\sqrt{R_4}\sqrt{\frac{g_m}{R_2g_m+1}}\sqrt{2R_3+R_4}}{2C_2R_2R_3g_m+C_2R_2R_4g_m+2C_2R_3+C_2R_4+2C_4R_3R_4g_m}$
wo: $\sqrt{\frac{2R_3g_m+R_4g_m}{2C_2C_4R_2R_3R_4g_m+2C_2C_4R_3R_4}}$
bandwidth: $\frac{\sqrt{\frac{2R_3g_m+R_4g_m}{2C_2C_4R_2R_3R_4g_m+2C_2C_4R_3R_4}}\left(2C_2R_2R_3g_m+C_2R_2R_4g_m+2C_2R_3+C_2R_4+2C_4R_3R_4g_m\right)}{\sqrt{2}\sqrt{C_2}\sqrt{C_4}R_2\sqrt{R_3}\sqrt{R_4}g_m\sqrt{\frac{g_m}{R_2g_m+1}}\sqrt{2R_3+R_4}+\sqrt{2}\sqrt{C_2}\sqrt{C_4}\sqrt{R_3}\sqrt{R_4}\sqrt{\frac{g_m}{R_2g_m+1}}\sqrt{2R_3+R_4}}$
K-LP: $\frac{R_3R_4}{2R_3+R_4}$
K-HP: 0
K-BP: $\frac{C_2R_2R_3R_4g_m+C_2R_3R_4}{2C_2R_2R_3g_m+C_2R_2R_4g_m+2C_2R_3+C_2R_4+2C_4R_3R_4g_m}$
Qz: None
Wz: None

8.19 X-INVALID-NUMER-19 $Z(s) = \left(\infty,\ R_2 + \frac{1}{C_2 s},\ \frac{1}{C_3 s},\ R_4,\ \infty,\ \infty\right)$

$$H(s) = \frac{R_4g_m + s\left(C_2R_2R_4g_m + C_2R_4\right)}{2g_m + s^2\left(C_2C_3R_2R_4g_m + C_2C_3R_4\right) + s\left(2C_2R_2g_m + 2C_2 + C_3R_4g_m\right)}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_2}\sqrt{C_3}R_2\sqrt{R_4}g_m\sqrt{\frac{g_m}{R_2g_m+1}}+\sqrt{2}\sqrt{C_2}\sqrt{C_3}\sqrt{R_4}\sqrt{\frac{g_m}{R_2g_m+1}}}{2C_2R_2g_m+2C_2+C_3R_4g_m}$
wo: $\sqrt{2}\sqrt{\frac{g_m}{C_2C_3R_2R_4g_m+C_2C_3R_4}}$
bandwidth: $\frac{\sqrt{2}\sqrt{\frac{g_m}{C_2C_3R_2R_4g_m+C_2C_3R_4}}\left(2C_2R_2g_m+2C_2+C_3R_4g_m\right)}{\sqrt{2}\sqrt{C_2}\sqrt{C_3}R_2\sqrt{R_4}g_m\sqrt{\frac{g_m}{R_2g_m+1}}+\sqrt{2}\sqrt{C_2}\sqrt{C_3}\sqrt{R_4}\sqrt{\frac{g_m}{R_2g_m+1}}}$
K-LP: $\frac{R_4}{2}$
K-HP: 0
K-BP: $\frac{C_2R_2R_4g_m+C_2R_4}{2C_2R_2g_m+2C_2+C_3R_4g_m}$
Qz: None
Wz: None

8.20 X-INVALID-NUMER-20 $Z(s) = \left(\infty,\ R_2 + \frac{1}{C_2 s},\ \frac{1}{C_3 s},\ \frac{R_4}{C_4 R_4 s+1},\ \infty,\ \infty\right)$

$$H(s) = \frac{R_4 g_m + s\left(C_2 R_2 R_4 g_m + C_2 R_4\right)}{2 g_m + s^2\left(C_2 C_3 R_2 R_4 g_m + C_2 C_3 R_4 + 2 C_2 C_4 R_2 R_4 g_m + 2 C_2 C_4 R_4\right) + s\left(2 C_2 R_2 g_m + 2 C_2 + C_3 R_4 g_m + 2 C_4 R_4 g_m\right)}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_2}C_3R_2\sqrt{R_4}g_m\sqrt{\frac{g_m}{C_3R_2g_m+C_3+2C_4R_2g_m+2C_4}}+\sqrt{2}\sqrt{C_2}C_3\sqrt{R_4}\sqrt{\frac{g_m}{C_3R_2g_m+C_3+2C_4R_2g_m+2C_4}}+2\sqrt{2}\sqrt{C_2}C_4R_2\sqrt{R_4}g_m\sqrt{\frac{g_m}{C_3R_2g_m+C_3+2C_4R_2g_m+2C_4}}+2\sqrt{2}\sqrt{C_2}C_4\sqrt{R_4}\sqrt{\frac{g_m}{C_3R_2g_m+C_3+2C_4R_2g_m+2C_4}}}{2C_2R_2g_m+2C_2+C_3R_4g_m+2C_4R_4g_m}$

wo: $\sqrt{2}\sqrt{\frac{g_m}{C_2C_3R_2R_4g_m+C_2C_3R_4+2C_2C_4R_2R_4g_m+2C_2C_4R_4}}$

bandwidth: $\frac{\sqrt{2}\sqrt{\frac{g_m}{C_2C_3R_2R_4g_m+C_2C_3R_4+2C_2C_4R_2R_4g_m+2C_2C_4R_4}}(2C_2R_2g_m+2C_2+C_3R_4g_m+2C_4R_4g_m)}{\sqrt{2}\sqrt{C_2}C_3R_2\sqrt{R_4}g_m\sqrt{\frac{g_m}{C_3R_2g_m+C_3+2C_4R_2g_m+2C_4}}+\sqrt{2}\sqrt{C_2}C_3\sqrt{R_4}\sqrt{\frac{g_m}{C_3R_2g_m+C_3+2C_4R_2g_m+2C_4}}+2\sqrt{2}\sqrt{C_2}C_4R_2\sqrt{R_4}g_m\sqrt{\frac{g_m}{C_3R_2g_m+C_3+2C_4R_2g_m+2C_4}}+2\sqrt{2}\sqrt{C_2}C_4\sqrt{R_4}\sqrt{\frac{g_m}{C_3R_2g_m+C_3+2C_4R_2g_m+2C_4}}}$

K-LP: $\frac{R_4}{2}$

K-HP: 0

K-BP: $\frac{C_2R_2R_4g_m+C_2R_4}{2C_2R_2g_m+2C_2+C_3R_4g_m+2C_4R_4g_m}$

Qz: None

Wz: None

8.21 X-INVALID-NUMER-21 $Z(s) = \left(\infty,\ R_2 + \frac{1}{C_2 s},\ \frac{R_3}{C_3 R_3 s+1},\ R_4,\ \infty,\ \infty\right)$

$$H(s) = \frac{R_3 R_4 g_m + s\left(C_2 R_2 R_3 R_4 g_m + C_2 R_3 R_4\right)}{2 R_3 g_m + R_4 g_m + s^2\left(C_2 C_3 R_2 R_3 R_4 g_m + C_2 C_3 R_3 R_4\right) + s\left(2 C_2 R_2 R_3 g_m + C_2 R_2 R_4 g_m + 2 C_2 R_3 + C_2 R_4 + C_3 R_3 R_4 g_m\right)}$$

Parameters:

Q: $\frac{\sqrt{C_2}\sqrt{C_3}R_2\sqrt{R_3}\sqrt{R_4}g_m\sqrt{\frac{g_m}{R_2g_m+1}}\sqrt{2R_3+R_4}+\sqrt{C_2}\sqrt{C_3}\sqrt{R_3}\sqrt{R_4}\sqrt{\frac{g_m}{R_2g_m+1}}\sqrt{2R_3+R_4}}{2C_2R_2R_3g_m+C_2R_2R_4g_m+2C_2R_3+C_2R_4+C_3R_3R_4g_m}$

wo: $\sqrt{\frac{2R_3g_m+R_4g_m}{C_2C_3R_2R_3R_4g_m+C_2C_3R_3R_4}}$

bandwidth: $\frac{\sqrt{\frac{2R_3g_m+R_4g_m}{C_2C_3R_2R_3R_4g_m+C_2C_3R_3R_4}}(2C_2R_2R_3g_m+C_2R_2R_4g_m+2C_2R_3+C_2R_4+C_3R_3R_4g_m)}{\sqrt{C_2}\sqrt{C_3}R_2\sqrt{R_3}\sqrt{R_4}g_m\sqrt{\frac{g_m}{R_2g_m+1}}\sqrt{2R_3+R_4}+\sqrt{C_2}\sqrt{C_3}\sqrt{R_3}\sqrt{R_4}\sqrt{\frac{g_m}{R_2g_m+1}}\sqrt{2R_3+R_4}}$

K-LP: $\frac{R_3R_4}{2R_3+R_4}$

K-HP: 0

K-BP: $\frac{C_2R_2R_3R_4g_m+C_2R_3R_4}{2C_2R_2R_3g_m+C_2R_2R_4g_m+2C_2R_3+C_2R_4+C_3R_3R_4g_m}$

Qz: None

Wz: None

8.22 X-INVALID-NUMER-22 $Z(s) = \left(\infty,\ R_2 + \frac{1}{C_2 s},\ \frac{R_3}{C_3 R_3 s+1},\ \frac{1}{C_4 s},\ \infty,\ \infty\right)$

$$H(s) = \frac{R_3 g_m + s\left(C_2 R_2 R_3 g_m + C_2 R_3\right)}{g_m + s^2\left(C_2 C_3 R_2 R_3 g_m + C_2 C_3 R_3 + 2 C_2 C_4 R_2 R_3 g_m + 2 C_2 C_4 R_3\right) + s\left(C_2 R_2 g_m + C_2 + C_3 R_3 g_m + 2 C_4 R_3 g_m\right)}$$

Parameters:

Q: $\frac{\sqrt{C_2}C_3R_2\sqrt{R_3}g_m\sqrt{\frac{g_m}{C_3R_2g_m+C_3+2C_4R_2g_m+2C_4}}+\sqrt{C_2}C_3\sqrt{R_3}\sqrt{\frac{g_m}{C_3R_2g_m+C_3+2C_4R_2g_m+2C_4}}+2\sqrt{C_2}C_4R_2\sqrt{R_3}g_m\sqrt{\frac{g_m}{C_3R_2g_m+C_3+2C_4R_2g_m+2C_4}}+2\sqrt{C_2}C_4\sqrt{R_3}\sqrt{\frac{g_m}{C_3R_2g_m+C_3+2C_4R_2g_m+2C_4}}}{C_2R_2g_m+C_2+C_3R_3g_m+2C_4R_3g_m}$

wo: $\sqrt{\frac{g_m}{C_2C_3R_2R_3g_m+C_2C_3R_3+2C_2C_4R_2R_3g_m+2C_2C_4R_3}}$

bandwidth: $\frac{\sqrt{\frac{g_m}{C_2C_3R_2R_3g_m+C_2C_3R_3+2C_2C_4R_2R_3g_m+2C_2C_4R_3}}(C_2R_2g_m+C_2+C_3R_3g_m+2C_4R_3g_m)}{\sqrt{C_2}C_3R_2\sqrt{R_3}g_m\sqrt{\frac{g_m}{C_3R_2g_m+C_3+2C_4R_2g_m+2C_4}}+\sqrt{C_2}C_3\sqrt{R_3}\sqrt{\frac{g_m}{C_3R_2g_m+C_3+2C_4R_2g_m+2C_4}}+2\sqrt{C_2}C_4R_2\sqrt{R_3}g_m\sqrt{\frac{g_m}{C_3R_2g_m+C_3+2C_4R_2g_m+2C_4}}+2\sqrt{C_2}C_4\sqrt{R_3}\sqrt{\frac{g_m}{C_3R_2g_m+C_3+2C_4R_2g_m+2C_4}}}$

K-LP: R_3

K-HP: 0

K-BP: $\frac{C_2R_2R_3g_m+C_2R_3}{C_2R_2g_m+C_2+C_3R_3g_m+2C_4R_3g_m}$

Qz: None

Wz: None

8.23 X-INVALID-NUMER-23 $Z(s) = \left(\infty,\ R_2 + \frac{1}{C_2 s},\ \frac{R_3}{C_3 R_3 s+1},\ \frac{R_4}{C_4 R_4 s+1},\ \infty,\ \infty\right)$

$$H(s) = \frac{R_3 R_4 g_m + s\left(C_2 R_2 R_3 R_4 g_m + C_2 R_3 R_4\right)}{2R_3 g_m + R_4 g_m + s^2\left(C_2 C_3 R_2 R_3 R_4 g_m + C_2 C_3 R_3 R_4 + 2C_2 C_4 R_2 R_3 R_4 g_m + 2C_2 C_4 R_3 R_4\right) + s\left(2C_2 R_2 R_3 g_m + C_2 R_2 R_4 g_m + 2C_2 R_3 + C_2 R_4 + C_3 R_3 R_4 g_m + 2C_4 R_3 R_4 g_m\right)}$$

Parameters:

Q: $\frac{\sqrt{C_2} C_3 R_2 \sqrt{R_3} \sqrt{R_4} g_m \sqrt{\frac{g_m}{C_3 R_2 g_m + C_3 + 2C_4 R_2 g_m + 2C_4}} \sqrt{2R_3+R_4} + \sqrt{C_2} C_3 \sqrt{R_3} \sqrt{R_4} \sqrt{\frac{g_m}{C_3 R_2 g_m + C_3 + 2C_4 R_2 g_m + 2C_4}} \sqrt{2R_3+R_4} + 2\sqrt{C_2} C_4 R_2 \sqrt{R_3} \sqrt{R_4} g_m \sqrt{\frac{g_m}{C_3 R_2 g_m + C_3 + 2C_4 R_2 g_m + 2C_4}} \sqrt{2R_3+R_4} + 2\sqrt{C_2} C_4 \sqrt{R_3} \sqrt{R_4} \sqrt{\frac{g_m}{C_3 R_2 g_m + C_3 + 2C_4 R_2 g_m + 2C_4}} \sqrt{2R_3+R_4}}{2C_2 R_2 R_3 g_m + C_2 R_2 R_4 g_m + 2C_2 R_3 + C_2 R_4 + C_3 R_3 R_4 g_m + 2C_4 R_3 R_4 g_m}$

wo: $\sqrt{\frac{2R_3 g_m + R_4 g_m}{C_2 C_3 R_2 R_3 R_4 g_m + C_2 C_3 R_3 R_4 + 2C_2 C_4 R_2 R_3 R_4 g_m + 2C_2 C_4 R_3 R_4}}$

bandwidth: $\frac{\sqrt{\frac{2R_3 g_m + R_4 g_m}{C_2 C_3 R_2 R_3 R_4 g_m + C_2 C_3 R_3 R_4 + 2C_2 C_4 R_2 R_3 R_4 g_m + 2C_2 C_4 R_3 R_4}} \left(2C_2 R_2 R_3 g_m + C_2 R_2 R_4 g_m + 2C_2 R_3 + C_2 R_4 + C_3 R_3 R_4 g_m + 2C_4 R_3 R_4 g_m\right)}{\sqrt{C_2} C_3 R_2 \sqrt{R_3} \sqrt{R_4} g_m \sqrt{\frac{g_m}{C_3 R_2 g_m + C_3 + 2C_4 R_2 g_m + 2C_4}} \sqrt{2R_3+R_4} + \sqrt{C_2} C_3 \sqrt{R_3} \sqrt{R_4} \sqrt{\frac{g_m}{C_3 R_2 g_m + C_3 + 2C_4 R_2 g_m + 2C_4}} \sqrt{2R_3+R_4} + 2\sqrt{C_2} C_4 R_2 \sqrt{R_3} \sqrt{R_4} g_m \sqrt{\frac{g_m}{C_3 R_2 g_m + C_3 + 2C_4 R_2 g_m + 2C_4}} \sqrt{2R_3+R_4} + 2\sqrt{C_2} C_4 \sqrt{R_3} \sqrt{R_4} \sqrt{\frac{g_m}{C_3 R_2 g_m + C_3 + 2C_4 R_2 g_m + 2C_4}} \sqrt{2R_3+R_4}}$

K-LP: $\frac{R_3 R_4}{2R_3 + R_4}$

K-HP: 0

K-BP: $\frac{C_2 R_2 R_3 R_4 g_m + C_2 R_3 R_4}{2C_2 R_2 R_3 g_m + C_2 R_2 R_4 g_m + 2C_2 R_3 + C_2 R_4 + C_3 R_3 R_4 g_m + 2C_4 R_3 R_4 g_m}$

Qz: None

Wz: None

9 X-INVALID-ORDER

9.1 X-INVALID-ORDER-1 $Z(s) = \left(\infty,\ R_2,\ R_3,\ R_4,\ \infty,\ \infty\right)$

$$H(s) = \frac{R_2 R_3 R_4 g_m + R_3 R_4}{2R_2 R_3 g_m + R_2 R_4 g_m + 2R_3 + R_4}$$

9.2 X-INVALID-ORDER-2 $Z(s) = \left(\infty,\ R_2,\ R_3,\ \frac{1}{C_4 s},\ \infty,\ \infty\right)$

$$H(s) = \frac{R_2 R_3 g_m + R_3}{R_2 g_m + s\left(2C_4 R_2 R_3 g_m + 2C_4 R_3\right) + 1}$$

9.3 X-INVALID-ORDER-3 $Z(s) = \left(\infty,\ R_2,\ R_3,\ \frac{R_4}{C_4 R_4 s+1},\ \infty,\ \infty\right)$

$$H(s) = \frac{R_2 R_3 R_4 g_m + R_3 R_4}{2R_2 R_3 g_m + R_2 R_4 g_m + 2R_3 + R_4 + s\left(2C_4 R_2 R_3 R_4 g_m + 2C_4 R_3 R_4\right)}$$

9.4 X-INVALID-ORDER-4 $Z(s) = \left(\infty,\ R_2,\ R_3,\ R_4 + \frac{1}{C_4 s},\ \infty,\ \infty\right)$

$$H(s) = \frac{R_2 R_3 g_m + R_3 + s\left(C_4 R_2 R_3 R_4 g_m + C_4 R_3 R_4\right)}{R_2 g_m + s\left(2C_4 R_2 R_3 g_m + C_4 R_2 R_4 g_m + 2C_4 R_3 + C_4 R_4\right) + 1}$$

9.5 X-INVALID-ORDER-5 $Z(s) = \left(\infty,\ R_2,\ \frac{1}{C_3 s},\ R_4,\ \infty,\ \infty\right)$

$$H(s) = \frac{R_2 R_4 g_m + R_4}{2R_2 g_m + s\left(C_3 R_2 R_4 g_m + C_3 R_4\right) + 2}$$

9.6 X-INVALID-ORDER-6 $Z(s) = \left(\infty,\ R_2,\ \frac{1}{C_3 s},\ \frac{1}{C_4 s},\ \infty,\ \infty\right)$

$$H(s) = \frac{R_2 g_m + 1}{s\left(C_3 R_2 g_m + C_3 + 2C_4 R_2 g_m + 2C_4\right)}$$

9.7 X-INVALID-ORDER-7 $Z(s) = \left(\infty,\ R_2,\ \frac{1}{C_3 s},\ \frac{R_4}{C_4 R_4 s+1},\ \infty,\ \infty\right)$

$$H(s) = \frac{R_2 R_4 g_m + R_4}{2R_2 g_m + s\left(C_3 R_2 R_4 g_m + C_3 R_4 + 2C_4 R_2 R_4 g_m + 2C_4 R_4\right) + 2}$$

9.8 X-INVALID-ORDER-8 $Z(s) = \left(\infty,\ R_2,\ \frac{1}{C_3 s},\ R_4 + \frac{1}{C_4 s},\ \infty,\ \infty\right)$

$$H(s) = \frac{R_2 g_m + s\left(C_4 R_2 R_4 g_m + C_4 R_4\right) + 1}{s^2\left(C_3 C_4 R_2 R_4 g_m + C_3 C_4 R_4\right) + s\left(C_3 R_2 g_m + C_3 + 2C_4 R_2 g_m + 2C_4\right)}$$

9.9 X-INVALID-ORDER-9 $Z(s) = \left(\infty,\ R_2,\ \frac{R_3}{C_3 R_3 s+1},\ R_4,\ \infty,\ \infty\right)$

$$H(s) = \frac{R_2 R_3 R_4 g_m + R_3 R_4}{2R_2 R_3 g_m + R_2 R_4 g_m + 2R_3 + R_4 + s\left(C_3 R_2 R_3 R_4 g_m + C_3 R_3 R_4\right)}$$

9.10 X-INVALID-ORDER-10 $Z(s) = \left(\infty,\ R_2,\ \frac{R_3}{C_3 R_3 s+1},\ \frac{1}{C_4 s},\ \infty,\ \infty\right)$

$$H(s) = \frac{R_2 R_3 g_m + R_3}{R_2 g_m + s\left(C_3 R_2 R_3 g_m + C_3 R_3 + 2C_4 R_2 R_3 g_m + 2C_4 R_3\right) + 1}$$

9.11 X-INVALID-ORDER-11 $Z(s) = \left(\infty,\ R_2,\ \frac{R_3}{C_3 R_3 s+1},\ \frac{R_4}{C_4 R_4 s+1},\ \infty,\ \infty\right)$

$$H(s) = \frac{R_2 R_3 R_4 g_m + R_3 R_4}{2R_2 R_3 g_m + R_2 R_4 g_m + 2R_3 + R_4 + s\left(C_3 R_2 R_3 R_4 g_m + C_3 R_3 R_4 + 2C_4 R_2 R_3 R_4 g_m + 2C_4 R_3 R_4\right)}$$

9.12 X-INVALID-ORDER-12 $Z(s) = \left(\infty,\ R_2,\ R_3 + \frac{1}{C_3 s},\ R_4,\ \infty,\ \infty\right)$

$$H(s) = \frac{R_2 R_4 g_m + R_4 + s\left(C_3 R_2 R_3 R_4 g_m + C_3 R_3 R_4\right)}{2R_2 g_m + s\left(2C_3 R_2 R_3 g_m + C_3 R_2 R_4 g_m + 2C_3 R_3 + C_3 R_4\right) + 2}$$

9.13 X-INVALID-ORDER-13 $Z(s) = \left(\infty,\ R_2,\ R_3 + \frac{1}{C_3 s},\ \frac{1}{C_4 s},\ \infty,\ \infty\right)$

$$H(s) = \frac{R_2 g_m + s\left(C_3 R_2 R_3 g_m + C_3 R_3\right) + 1}{s^2\left(2C_3 C_4 R_2 R_3 g_m + 2C_3 C_4 R_3\right) + s\left(C_3 R_2 g_m + C_3 + 2C_4 R_2 g_m + 2C_4\right)}$$

9.14 X-INVALID-ORDER-14 $Z(s) = \left(\infty,\ R_2,\ R_3 + \frac{1}{C_3 s},\ R_4 + \frac{1}{C_4 s},\ \infty,\ \infty\right)$

$$H(s) = \frac{R_2 g_m + s^2\left(C_3 C_4 R_2 R_3 R_4 g_m + C_3 C_4 R_3 R_4\right) + s\left(C_3 R_2 R_3 g_m + C_3 R_3 + C_4 R_2 R_4 g_m + C_4 R_4\right) + 1}{s^2\left(2C_3 C_4 R_2 R_3 g_m + C_3 C_4 R_2 R_4 g_m + 2C_3 C_4 R_3 + C_3 C_4 R_4\right) + s\left(C_3 R_2 g_m + C_3 + 2C_4 R_2 g_m + 2C_4\right)}$$

9.15 X-INVALID-ORDER-15 $Z(s) = \left(\infty,\ \frac{1}{C_2 s},\ R_3,\ R_4,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_2 R_3 R_4 s + R_3 R_4 g_m}{2R_3 g_m + R_4 g_m + s\left(2C_2 R_3 + C_2 R_4\right)}$$

9.16 X-INVALID-ORDER-16 $Z(s) = \left(\infty,\ \frac{1}{C_2 s},\ \frac{1}{C_3 s},\ \frac{1}{C_4 s},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_2 s + g_m}{s^2\left(C_2 C_3 + 2C_2 C_4\right) + s\left(C_3 g_m + 2C_4 g_m\right)}$$

9.17 X-INVALID-ORDER-17 $Z(s) = \left(\infty,\ \frac{1}{C_2 s},\ \frac{1}{C_3 s},\ R_4 + \frac{1}{C_4 s},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_2C_4R_4s^2 + g_m + s\left(C_2 + C_4R_4g_m\right)}{C_2C_3C_4R_4s^3 + s^2\left(C_2C_3 + 2C_2C_4 + C_3C_4R_4g_m\right) + s\left(C_3g_m + 2C_4g_m\right)}$$

9.18 X-INVALID-ORDER-18 $Z(s) = \left(\infty,\ \frac{1}{C_2 s},\ \frac{R_3}{C_3R_3s+1},\ R_4 + \frac{1}{C_4 s},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_2C_4R_3R_4s^2 + R_3g_m + s\left(C_2R_3 + C_4R_3R_4g_m\right)}{C_2C_3C_4R_3R_4s^3 + g_m + s^2\left(C_2C_3R_3 + 2C_2C_4R_3 + C_2C_4R_4 + C_3C_4R_3R_4g_m\right) + s\left(C_2 + C_3R_3g_m + 2C_4R_3g_m + C_4R_4g_m\right)}$$

9.19 X-INVALID-ORDER-19 $Z(s) = \left(\infty,\ \frac{1}{C_2 s},\ R_3 + \frac{1}{C_3 s},\ \frac{1}{C_4 s},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_2C_3R_3s^2 + g_m + s\left(C_2 + C_3R_3g_m\right)}{2C_2C_3C_4R_3s^3 + s^2\left(C_2C_3 + 2C_2C_4 + 2C_3C_4R_3g_m\right) + s\left(C_3g_m + 2C_4g_m\right)}$$

9.20 X-INVALID-ORDER-20 $Z(s) = \left(\infty,\ \frac{1}{C_2 s},\ R_3 + \frac{1}{C_3 s},\ \frac{R_4}{C_4R_4s+1},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_2C_3R_3R_4s^2 + R_4g_m + s\left(C_2R_4 + C_3R_3R_4g_m\right)}{2C_2C_3C_4R_3R_4s^3 + 2g_m + s^2\left(2C_2C_3R_3 + C_2C_3R_4 + 2C_2C_4R_4 + 2C_3C_4R_3R_4g_m\right) + s\left(2C_2 + 2C_3R_3g_m + C_3R_4g_m + 2C_4R_4g_m\right)}$$

9.21 X-INVALID-ORDER-21 $Z(s) = \left(\infty,\ \frac{1}{C_2 s},\ R_3 + \frac{1}{C_3 s},\ R_4 + \frac{1}{C_4 s},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_2C_3C_4R_3R_4s^3 + g_m + s^2\left(C_2C_3R_3 + C_2C_4R_4 + C_3C_4R_3R_4g_m\right) + s\left(C_2 + C_3R_3g_m + C_4R_4g_m\right)}{s^3\left(2C_2C_3C_4R_3 + C_2C_3C_4R_4\right) + s^2\left(C_2C_3 + 2C_2C_4 + 2C_3C_4R_3g_m + C_3C_4R_4g_m\right) + s\left(C_3g_m + 2C_4g_m\right)}$$

9.22 X-INVALID-ORDER-22 $Z(s) = \left(\infty,\ \frac{R_2}{C_2R_2s+1},\ R_3,\ R_4,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_2R_2R_3R_4s + R_2R_3R_4g_m + R_3R_4}{2R_2R_3g_m + R_2R_4g_m + 2R_3 + R_4 + s\left(2C_2R_2R_3 + C_2R_2R_4\right)}$$

9.23 X-INVALID-ORDER-23 $Z(s) = \left(\infty,\ \frac{R_2}{C_2R_2s+1},\ \frac{1}{C_3 s},\ \frac{1}{C_4 s},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_2R_2s + R_2g_m + 1}{s^2\left(C_2C_3R_2 + 2C_2C_4R_2\right) + s\left(C_3R_2g_m + C_3 + 2C_4R_2g_m + 2C_4\right)}$$

9.24 X-INVALID-ORDER-24 $Z(s) = \left(\infty,\ \frac{R_2}{C_2R_2s+1},\ \frac{1}{C_3 s},\ R_4 + \frac{1}{C_4 s},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_2C_4R_2R_4s^2 + R_2g_m + s\left(C_2R_2 + C_4R_2R_4g_m + C_4R_4\right) + 1}{C_2C_3C_4R_2R_4s^3 + s^2\left(C_2C_3R_2 + 2C_2C_4R_2 + C_3C_4R_2R_4g_m + C_3C_4R_4\right) + s\left(C_3R_2g_m + C_3 + 2C_4R_2g_m + 2C_4\right)}$$

9.25 X-INVALID-ORDER-25 $Z(s) = \left(\infty,\ \frac{R_2}{C_2R_2s+1},\ \frac{R_3}{C_3R_3s+1},\ R_4 + \frac{1}{C_4 s},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_2C_4R_2R_3R_4s^2 + R_2R_3g_m + R_3 + s\left(C_2R_2R_3 + C_4R_2R_3R_4g_m + C_4R_3R_4\right)}{C_2C_3C_4R_2R_3R_4s^3 + R_2g_m + s^2\left(C_2C_3R_2R_3 + 2C_2C_4R_2R_3 + C_2C_4R_2R_4 + C_3C_4R_2R_3R_4g_m + C_3C_4R_3R_4\right) + s\left(C_2R_2 + C_3R_2R_3g_m + C_3R_3 + 2C_4R_2R_3g_m + C_4R_2R_4g_m + 2C_4R_3 + C_4R_4\right) + 1}$$

9.26 X-INVALID-ORDER-26 $Z(s) = \left(\infty,\ \frac{R_2}{C_2R_2s+1},\ R_3 + \frac{1}{C_3 s},\ \frac{1}{C_4 s},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_2C_3R_2R_3s^2 + R_2g_m + s\left(C_2R_2 + C_3R_2R_3g_m + C_3R_3\right) + 1}{2C_2C_3C_4R_2R_3s^3 + s^2\left(C_2C_3R_2 + 2C_2C_4R_2 + 2C_3C_4R_2R_3g_m + 2C_3C_4R_3\right) + s\left(C_3R_2g_m + C_3 + 2C_4R_2g_m + 2C_4\right)}$$

9.27 X-INVALID-ORDER-27 $Z(s) = \left(\infty,\ \frac{R_2}{C_2R_2s+1},\ R_3 + \frac{1}{C_3s},\ \frac{R_4}{C_4R_4s+1},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_2C_3R_2R_3R_4s^2 + R_2R_4g_m + R_4 + s\left(C_2R_2R_4 + C_3R_2R_3R_4g_m + C_3R_3R_4\right)}{2C_2C_3C_4R_2R_3R_4s^3 + 2R_2g_m + s^2\left(2C_2C_3R_2R_3 + C_2C_3R_2R_4 + 2C_2C_4R_2R_4 + 2C_3C_4R_2R_3R_4g_m + 2C_3C_4R_3R_4\right) + s\left(2C_2R_2 + 2C_3R_2R_3g_m + C_3R_2R_4g_m + 2C_3R_3 + C_3R_4 + 2C_4R_2R_4g_m + 2C_4R_4\right) + 2}$$

9.28 X-INVALID-ORDER-28 $Z(s) = \left(\infty,\ \frac{R_2}{C_2R_2s+1},\ R_3 + \frac{1}{C_3s},\ R_4 + \frac{1}{C_4s},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_2C_3C_4R_2R_3R_4s^3 + R_2g_m + s^2\left(C_2C_3R_2R_3 + C_2C_4R_2R_4 + C_3C_4R_2R_3R_4g_m + C_3C_4R_3R_4\right) + s\left(C_2R_2 + C_3R_2R_3g_m + C_3R_3 + C_4R_2R_4g_m + C_4R_4\right) + 1}{s^3\left(2C_2C_3C_4R_2R_3 + C_2C_3C_4R_2R_4\right) + s^2\left(C_2C_3R_2 + 2C_2C_4R_2 + 2C_3C_4R_2R_3g_m + C_3C_4R_2R_4g_m + 2C_3C_4R_3 + C_3C_4R_4\right) + s\left(C_3R_2g_m + C_3 + 2C_4R_2g_m + 2C_4\right)}$$

9.29 X-INVALID-ORDER-29 $Z(s) = \left(\infty,\ R_2 + \frac{1}{C_2s},\ R_3,\ R_4,\ \infty,\ \infty\right)$

$$H(s) = \frac{R_3R_4g_m + s\left(C_2R_2R_3R_4g_m + C_2R_3R_4\right)}{2R_3g_m + R_4g_m + s\left(2C_2R_2R_3g_m + C_2R_2R_4g_m + 2C_2R_3 + C_2R_4\right)}$$

9.30 X-INVALID-ORDER-30 $Z(s) = \left(\infty,\ R_2 + \frac{1}{C_2s},\ \frac{1}{C_3s},\ \frac{1}{C_4s},\ \infty,\ \infty\right)$

$$H(s) = \frac{g_m + s\left(C_2R_2g_m + C_2\right)}{s^2\left(C_2C_3R_2g_m + C_2C_3 + 2C_2C_4R_2g_m + 2C_2C_4\right) + s\left(C_3g_m + 2C_4g_m\right)}$$

9.31 X-INVALID-ORDER-31 $Z(s) = \left(\infty,\ R_2 + \frac{1}{C_2s},\ \frac{1}{C_3s},\ R_4 + \frac{1}{C_4s},\ \infty,\ \infty\right)$

$$H(s) = \frac{g_m + s^2\left(C_2C_4R_2R_4g_m + C_2C_4R_4\right) + s\left(C_2R_2g_m + C_2 + C_4R_4g_m\right)}{s^3\left(C_2C_3C_4R_2R_4g_m + C_2C_3C_4R_4\right) + s^2\left(C_2C_3R_2g_m + C_2C_3 + 2C_2C_4R_2g_m + 2C_2C_4 + C_3C_4R_4g_m\right) + s\left(C_3g_m + 2C_4g_m\right)}$$

9.32 X-INVALID-ORDER-32 $Z(s) = \left(\infty,\ R_2 + \frac{1}{C_2s},\ \frac{R_3}{C_3R_3s+1},\ R_4 + \frac{1}{C_4s},\ \infty,\ \infty\right)$

$$H(s) = \frac{R_3g_m + s^2\left(C_2C_4R_2R_3R_4g_m + C_2C_4R_3R_4\right) + s\left(C_2R_2R_3g_m + C_2R_3 + C_4R_3R_4g_m\right)}{g_m + s^3\left(C_2C_3C_4R_2R_3R_4g_m + C_2C_3C_4R_3R_4\right) + s^2\left(C_2C_3R_2R_3g_m + C_2C_3R_3 + 2C_2C_4R_2R_3g_m + C_2C_4R_2R_4g_m + 2C_2C_4R_3 + C_2C_4R_4 + C_3C_4R_3R_4g_m\right) + s\left(C_2R_2g_m + C_2 + C_3R_3g_m + 2C_4R_3g_m + C_4R_4g_m\right)}$$

9.33 X-INVALID-ORDER-33 $Z(s) = \left(\infty,\ R_2 + \frac{1}{C_2s},\ R_3 + \frac{1}{C_3s},\ \frac{1}{C_4s},\ \infty,\ \infty\right)$

$$H(s) = \frac{g_m + s^2\left(C_2C_3R_2R_3g_m + C_2C_3R_3\right) + s\left(C_2R_2g_m + C_2 + C_3R_3g_m\right)}{s^3\left(2C_2C_3C_4R_2R_3g_m + 2C_2C_3C_4R_3\right) + s^2\left(C_2C_3R_2g_m + C_2C_3 + 2C_2C_4R_2g_m + 2C_2C_4 + 2C_3C_4R_3g_m\right) + s\left(C_3g_m + 2C_4g_m\right)}$$

9.34 X-INVALID-ORDER-34 $Z(s) = \left(\infty,\ R_2 + \frac{1}{C_2s},\ R_3 + \frac{1}{C_3s},\ \frac{R_4}{C_4R_4s+1},\ \infty,\ \infty\right)$

$$H(s) = \frac{R_4g_m + s^2\left(C_2C_3R_2R_3R_4g_m + C_2C_3R_3R_4\right) + s\left(C_2R_2R_4g_m + C_2R_4 + C_3R_3R_4g_m\right)}{2g_m + s^3\left(2C_2C_3C_4R_2R_3R_4g_m + 2C_2C_3C_4R_3R_4\right) + s^2\left(2C_2C_3R_2R_3g_m + C_2C_3R_2R_4g_m + 2C_2C_3R_3 + C_2C_3R_4 + 2C_2C_4R_2R_4g_m + 2C_2C_4R_4 + 2C_3C_4R_3R_4g_m\right) + s\left(2C_2R_2g_m + 2C_2 + 2C_3R_3g_m + C_3R_4g_m + 2C_4R_4g_m\right)}$$

9.35 X-INVALID-ORDER-35 $Z(s) = \left(\infty,\ R_2 + \frac{1}{C_2s},\ R_3 + \frac{1}{C_3s},\ R_4 + \frac{1}{C_4s},\ \infty,\ \infty\right)$

$$H(s) = \frac{g_m + s^3\left(C_2C_3C_4R_2R_3R_4g_m + C_2C_3C_4R_3R_4\right) + s^2\left(C_2C_3R_2R_3g_m + C_2C_3R_3 + C_2C_4R_2R_4g_m + C_2C_4R_4 + C_3C_4R_3R_4g_m\right) + s\left(C_2R_2g_m + C_2 + C_3R_3g_m + C_4R_4g_m\right)}{s^3\left(2C_2C_3C_4R_2R_3g_m + C_2C_3C_4R_2R_4g_m + 2C_2C_3C_4R_3 + C_2C_3C_4R_4\right) + s^2\left(C_2C_3R_2g_m + C_2C_3 + 2C_2C_4R_2g_m + 2C_2C_4 + 2C_3C_4R_3g_m + C_3C_4R_4g_m\right) + s\left(C_3g_m + 2C_4g_m\right)}$$

10 X-INVALID-WZ

10.1 X-INVALID-WZ-1 $Z(s) = \left(\infty,\ \frac{1}{C_2 s},\ R_3,\ R_4 + \frac{1}{C_4 s},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_2 C_4 R_3 R_4 s^2 + R_3 g_m + s\left(C_2 R_3 + C_4 R_3 R_4 g_m\right)}{g_m + s^2\left(2 C_2 C_4 R_3 + C_2 C_4 R_4\right) + s\left(C_2 + 2 C_4 R_3 g_m + C_4 R_4 g_m\right)}$$

Parameters:

Q: $\frac{\sqrt{C_2}\sqrt{C_4}\sqrt{g_m}\sqrt{2R_3+R_4}}{C_2+2C_4R_3g_m+C_4R_4g_m}$
wo: $\frac{\sqrt{g_m}}{\sqrt{2C_2C_4R_3+C_2C_4R_4}}$
bandwidth: $\frac{C_2+2C_4R_3g_m+C_4R_4g_m}{\sqrt{C_2}\sqrt{C_4}\sqrt{2R_3+R_4}\sqrt{2C_2C_4R_3+C_2C_4R_4}}$
K-LP: R_3
K-HP: $\frac{R_3R_4}{2R_3+R_4}$
K-BP: $\frac{C_2R_3+C_4R_3R_4g_m}{C_2+2C_4R_3g_m+C_4R_4g_m}$
Qz: None
Wz: $\frac{\sqrt{g_m}}{\sqrt{C_2}\sqrt{C_4}\sqrt{R_4}}$

10.2 X-INVALID-WZ-2 $Z(s) = \left(\infty,\ \frac{1}{C_2 s},\ R_3 + \frac{1}{C_3 s},\ R_4,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_2 C_3 R_3 R_4 s^2 + R_4 g_m + s\left(C_2 R_4 + C_3 R_3 R_4 g_m\right)}{2 g_m + s^2\left(2 C_2 C_3 R_3 + C_2 C_3 R_4\right) + s\left(2 C_2 + 2 C_3 R_3 g_m + C_3 R_4 g_m\right)}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_2}\sqrt{C_3}\sqrt{g_m}\sqrt{2R_3+R_4}}{2C_2+2C_3R_3g_m+C_3R_4g_m}$
wo: $\frac{\sqrt{2}\sqrt{g_m}}{\sqrt{2C_2C_3R_3+C_2C_3R_4}}$
bandwidth: $\frac{2C_2+2C_3R_3g_m+C_3R_4g_m}{\sqrt{C_2}\sqrt{C_3}\sqrt{2R_3+R_4}\sqrt{2C_2C_3R_3+C_2C_3R_4}}$
K-LP: $\frac{R_4}{2}$
K-HP: $\frac{R_3R_4}{2R_3+R_4}$
K-BP: $\frac{C_2R_4+C_3R_3R_4g_m}{2C_2+2C_3R_3g_m+C_3R_4g_m}$
Qz: None
Wz: $\frac{\sqrt{g_m}}{\sqrt{C_2}\sqrt{C_3}\sqrt{R_3}}$

10.3 X-INVALID-WZ-3 $Z(s) = \left(\infty,\ \frac{R_2}{C_2 R_2 s+1},\ R_3,\ R_4 + \frac{1}{C_4 s},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_2 C_4 R_2 R_3 R_4 s^2 + R_2 R_3 g_m + R_3 + s\left(C_2 R_2 R_3 + C_4 R_2 R_3 R_4 g_m + C_4 R_3 R_4\right)}{R_2 g_m + s^2\left(2 C_2 C_4 R_2 R_3 + C_2 C_4 R_2 R_4\right) + s\left(C_2 R_2 + 2 C_4 R_2 R_3 g_m + C_4 R_2 R_4 g_m + 2 C_4 R_3 + C_4 R_4\right) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_2}\sqrt{C_4}\sqrt{R_2}\sqrt{2R_3+R_4}\sqrt{R_2g_m+1}}{C_2R_2+2C_4R_2R_3g_m+C_4R_2R_4g_m+2C_4R_3+C_4R_4}$
wo: $\frac{\sqrt{R_2g_m+1}}{\sqrt{2C_2C_4R_2R_3+C_2C_4R_2R_4}}$
bandwidth: $\frac{C_2R_2+2C_4R_2R_3g_m+C_4R_2R_4g_m+2C_4R_3+C_4R_4}{\sqrt{C_2}\sqrt{C_4}\sqrt{R_2}\sqrt{2R_3+R_4}\sqrt{2C_2C_4R_2R_3+C_2C_4R_2R_4}}$
K-LP: R_3
K-HP: $\frac{R_3R_4}{2R_3+R_4}$
K-BP: $\frac{C_2R_2R_3+C_4R_2R_3R_4g_m+C_4R_3R_4}{C_2R_2+2C_4R_2R_3g_m+C_4R_2R_4g_m+2C_4R_3+C_4R_4}$
Qz: None
Wz: $\frac{\sqrt{R_2g_m+1}}{\sqrt{C_2}\sqrt{C_4}\sqrt{R_2}\sqrt{R_4}}$

10.4 X-INVALID-WZ-4 $Z(s) = \left(\infty,\ \frac{R_2}{C_2 R_2 s+1},\ R_3 + \frac{1}{C_3 s},\ R_4,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_2 C_3 R_2 R_3 R_4 s^2 + R_2 R_4 g_m + R_4 + s\left(C_2 R_2 R_4 + C_3 R_2 R_3 R_4 g_m + C_3 R_3 R_4\right)}{2 R_2 g_m + s^2\left(2 C_2 C_3 R_2 R_3 + C_2 C_3 R_2 R_4\right) + s\left(2 C_2 R_2 + 2 C_3 R_2 R_3 g_m + C_3 R_2 R_4 g_m + 2 C_3 R_3 + C_3 R_4\right) + 2}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_2}\sqrt{C_3}\sqrt{R_2}\sqrt{2R_3+R_4}\sqrt{R_2g_m+1}}{2C_2R_2+2C_3R_2R_3g_m+C_3R_2R_4g_m+2C_3R_3+C_3R_4}$
wo: $\frac{\sqrt{2R_2g_m+2}}{\sqrt{2C_2C_3R_2R_3+C_2C_3R_2R_4}}$

bandwidth: $\frac{\sqrt{2}\sqrt{2R_2g_m+2}(2C_2R_2+2C_3R_2R_3g_m+C_3R_2R_4g_m+2C_3R_3+C_3R_4)}{2\sqrt{C_2}\sqrt{C_3}\sqrt{R_2}\sqrt{2R_3+R_4}\sqrt{R_2g_m+1}\sqrt{2C_2C_3R_2R_3+C_2C_3R_2R_4}}$
K-LP: $\frac{R_4}{2}$
K-HP: $\frac{R_3R_4}{2R_3+R_4}$
K-BP: $\frac{C_2R_2R_4+C_3R_2R_3R_4g_m+C_3R_3R_4}{2C_2R_2+2C_3R_2R_3g_m+C_3R_2R_4g_m+2C_3R_3+C_3R_4}$
Qz: None
Wz: $\frac{\sqrt{R_2g_m+1}}{\sqrt{C_2}\sqrt{C_3}\sqrt{R_2}\sqrt{R_3}}$

10.5 X-INVALID-WZ-5 $Z(s) = \left(\infty,\ R_2+\frac{1}{C_2s},\ R_3,\ R_4+\frac{1}{C_4s},\ \infty,\ \infty\right)$

$$H(s) = \frac{R_3g_m + s^2\left(C_2C_4R_2R_3R_4g_m + C_2C_4R_3R_4\right) + s\left(C_2R_2R_3g_m + C_2R_3 + C_4R_3R_4g_m\right)}{g_m + s^2\left(2C_2C_4R_2R_3g_m + C_2C_4R_2R_4g_m + 2C_2C_4R_3 + C_2C_4R_4\right) + s\left(C_2R_2g_m + C_2 + 2C_4R_3g_m + C_4R_4g_m\right)}$$

Parameters:

Q: $\frac{2\sqrt{C_2}\sqrt{C_4}R_2R_3g_m\sqrt{\frac{g_m}{2R_2R_3g_m+R_2R_4g_m+2R_3+R_4}}+\sqrt{C_2}\sqrt{C_4}R_2R_4g_m\sqrt{\frac{g_m}{2R_2R_3g_m+R_2R_4g_m+2R_3+R_4}}+2\sqrt{C_2}\sqrt{C_4}R_3\sqrt{\frac{g_m}{2R_2R_3g_m+R_2R_4g_m+2R_3+R_4}}+\sqrt{C_2}\sqrt{C_4}R_4\sqrt{\frac{g_m}{2R_2R_3g_m+R_2R_4g_m+2R_3+R_4}}}{C_2R_2g_m+C_2+2C_4R_3g_m+C_4R_4g_m}$
wo: $\sqrt{\frac{g_m}{2C_2C_4R_2R_3g_m+C_2C_4R_2R_4g_m+2C_2C_4R_3+C_2C_4R_4}}$
bandwidth: $\frac{\sqrt{\frac{g_m}{2C_2C_4R_2R_3g_m+C_2C_4R_2R_4g_m+2C_2C_4R_3+C_2C_4R_4}}(C_2R_2g_m+C_2+2C_4R_3g_m+C_4R_4g_m)}{2\sqrt{C_2}\sqrt{C_4}R_2R_3g_m\sqrt{\frac{g_m}{2R_2R_3g_m+R_2R_4g_m+2R_3+R_4}}+\sqrt{C_2}\sqrt{C_4}R_2R_4g_m\sqrt{\frac{g_m}{2R_2R_3g_m+R_2R_4g_m+2R_3+R_4}}+2\sqrt{C_2}\sqrt{C_4}R_3\sqrt{\frac{g_m}{2R_2R_3g_m+R_2R_4g_m+2R_3+R_4}}+\sqrt{C_2}\sqrt{C_4}R_4\sqrt{\frac{g_m}{2R_2R_3g_m+R_2R_4g_m+2R_3+R_4}}}$
K-LP: R_3
K-HP: $\frac{R_3R_4}{2R_3+R_4}$
K-BP: $\frac{C_2R_2R_3g_m+C_2R_3+C_4R_3R_4g_m}{C_2R_2g_m+C_2+2C_4R_3g_m+C_4R_4g_m}$
Qz: None
Wz: $\sqrt{\frac{g_m}{C_2C_4R_2R_4g_m+C_2C_4R_4}}$

10.6 X-INVALID-WZ-6 $Z(s) = \left(\infty,\ R_2+\frac{1}{C_2s},\ R_3+\frac{1}{C_3s},\ R_4,\ \infty,\ \infty\right)$

$$H(s) = \frac{R_4g_m + s^2\left(C_2C_3R_2R_3R_4g_m + C_2C_3R_3R_4\right) + s\left(C_2R_2R_4g_m + C_2R_4 + C_3R_3R_4g_m\right)}{2g_m + s^2\left(2C_2C_3R_2R_3g_m + C_2C_3R_2R_4g_m + 2C_2C_3R_3 + C_2C_3R_4\right) + s\left(2C_2R_2g_m + 2C_2 + 2C_3R_3g_m + C_3R_4g_m\right)}$$

Parameters:

Q: $\frac{2\sqrt{2}\sqrt{C_2}\sqrt{C_3}R_2R_3g_m\sqrt{\frac{g_m}{2R_2R_3g_m+R_2R_4g_m+2R_3+R_4}}+\sqrt{2}\sqrt{C_2}\sqrt{C_3}R_2R_4g_m\sqrt{\frac{g_m}{2R_2R_3g_m+R_2R_4g_m+2R_3+R_4}}+2\sqrt{2}\sqrt{C_2}\sqrt{C_3}R_3\sqrt{\frac{g_m}{2R_2R_3g_m+R_2R_4g_m+2R_3+R_4}}+\sqrt{2}\sqrt{C_2}\sqrt{C_3}R_4\sqrt{\frac{g_m}{2R_2R_3g_m+R_2R_4g_m+2R_3+R_4}}}{2C_2R_2g_m+2C_2+2C_3R_3g_m+C_3R_4g_m}$
wo: $\sqrt{2}\sqrt{\frac{g_m}{2C_2C_3R_2R_3g_m+C_2C_3R_2R_4g_m+2C_2C_3R_3+C_2C_3R_4}}$
bandwidth: $\frac{\sqrt{2}\sqrt{\frac{g_m}{2C_2C_3R_2R_3g_m+C_2C_3R_2R_4g_m+2C_2C_3R_3+C_2C_3R_4}}(2C_2R_2g_m+2C_2+2C_3R_3g_m+C_3R_4g_m)}{2\sqrt{2}\sqrt{C_2}\sqrt{C_3}R_2R_3g_m\sqrt{\frac{g_m}{2R_2R_3g_m+R_2R_4g_m+2R_3+R_4}}+\sqrt{2}\sqrt{C_2}\sqrt{C_3}R_2R_4g_m\sqrt{\frac{g_m}{2R_2R_3g_m+R_2R_4g_m+2R_3+R_4}}+2\sqrt{2}\sqrt{C_2}\sqrt{C_3}R_3\sqrt{\frac{g_m}{2R_2R_3g_m+R_2R_4g_m+2R_3+R_4}}+\sqrt{2}\sqrt{C_2}\sqrt{C_3}R_4\sqrt{\frac{g_m}{2R_2R_3g_m+R_2R_4g_m+2R_3+R_4}}}$
K-LP: $\frac{R_4}{2}$
K-HP: $\frac{R_3R_4}{2R_3+R_4}$
K-BP: $\frac{C_2R_2R_4g_m+C_2R_4+C_3R_3R_4g_m}{2C_2R_2g_m+2C_2+2C_3R_3g_m+C_3R_4g_m}$
Qz: None
Wz: $\sqrt{\frac{g_m}{C_2C_3R_2R_3g_m+C_2C_3R_3}}$

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